

Bilateral Multi-Behavior Modeling for Reciprocal Recommendation in Online Recruitment

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Abstract—Recent years have witnessed the rapid development of online recruitment platforms, which provide a convenient way for matching job seekers and recruiters. Indeed, this is a *reciprocal recommendation* problem which needs to consider the preferences of both job seekers and recruiters simultaneously. Existing studies mainly focus on building recommendation models based on text matching or collaborative filtering methods. However, we propose that these methods are limited and insufficient, since the abundant multi-typed bilateral behaviors among users have been largely ignored. Therefore, in this paper, we propose a novel Bilateral Multi-Behavior Modeling (BAMBOO) method for reciprocal recommendation in online recruitment, which can model the multi-typed interactions from *expectation perspective* and *competitiveness perspective*. Specifically, for the expectation perspective, we propose to format the historical behaviors of different users as bilateral multi-behavior sequences, and we utilize a transformer-based model to learn the representations of what the users *want* to obtain. For the competitiveness perspective, we propose to construct a bilateral interaction heterogeneous graph to describe the entire recruitment market, and further utilize a heterogeneous graph transformer-based model to learn the representations of what the users *can* obtain. Moreover, we utilize contrastive learning methods to enhance these two modules. Furthermore, we propose to decompose the matching probability into the product of two parts, and we train our model based on a multi-task learning strategy. We conduct both offline experiments on real-world datasets and online A/B test, and the experiment results validate the effectiveness of our BAMBOO model compared with several state-of-the-art baseline methods.

Index Terms—Reciprocal recommendation, online recruitment, multi-behavior modeling, heterogeneous graph learning.

I. INTRODUCTION

RECENT years have witnessed the rapid development of online recruitment [1], [2], [3]. According to the report from The Insight Partners,¹ the global online recruitment market size is expected to grow from \$29.29 billion in 2021 to \$47.31 billion by 2028. In typical online recruitment platforms, job seekers and recruiters can make a two-way selection, i.e., the job seekers can send applications to the recruiters for a conversation, while the recruiters can also actively find suitable candidates and send invitations to them. Fig. 1 shows the process of a job seeker actively searching for a job on an online recruitment platform. Due to the large number of users on these platforms, it is almost impossible for users to independently screen suitable candidates. Therefore, Recommendation Systems (RS) have become the core component for overcoming information explosion.

Along this line, considerable research efforts have been made in building RS for online recruitment [2], [3], [4], [5], which mainly follow text matching or collaborative filtering paradigm. However, these existing methods often fail to comprehensively and efficiently model user behaviors in recruitment online recruitment platforms, which limits them to achieve satisfactory performance in practical application. To be specific, the user behaviors in recruitment platforms have the following three characteristics: 1) *Dynamicity*. User preference can change dynamically in recruitment platforms. For example, recently graduated students may have limited awareness of the job market and could face setbacks due to overly optimistic expectations. However, upon experiencing failure, they may adapt their expectations and job search strategies. 2) *Diversity*. The job seeker-recruiter interactive behaviors are multi-typed in recruitment platforms. For example, a job seeker may first *neglect* a job position since he is not interested in it. Later, he may *apply* to another position, and the application will have different possible results, such as being *accepted* or *rejected*. 3) *Bidirectionality*. As we mentioned before, both job seekers and recruiters can find candidates actively, and a user may show different preferences in these bilateral behaviors. For example, a job seeker may only apply to some large companies as the active party, while he may still accept applications from some small companies as the passive party.

¹<https://www.theinsightpartners.com/reports/online-recruitment-market>

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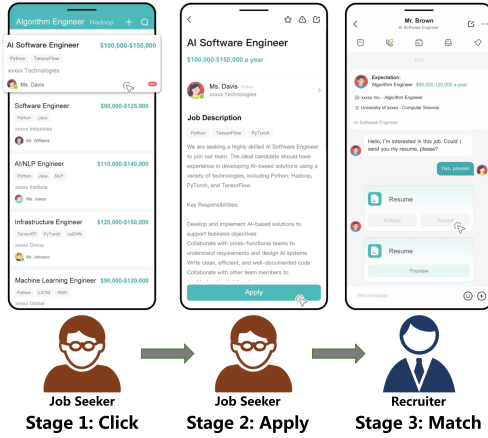


Fig. 1. An illustration of the process of a job seeker actively searching for a job on an online recruitment platform.

Indeed, the aforementioned characteristics bring the following challenges for building an online recruitment RS: 1) For a specific type of behavior, the information it contains is different when the active/passive party is different. To make things worse, we need to consider multiple types of behavior at the same time, which presents a challenge for our modeling. 2) Existing studies on behavior sequence modeling mainly focus on capturing the changes in user interests and expectations based on user historical records. However, The bidirectional selection mechanism makes all participants in the entire recruitment market connected as a competition network. Thus, we need to model the interactions from a higher-level to consider the market competition that exists between different recruiters and job seekers. 3) Different from traditional RS which only need to model the user preference for items, the RS for online recruitment platforms actually needs to model the features and preferences of both job seekers and recruiters, and recommend suitable candidates with high matching probabilities for the two stakeholders. In other words, we need a Reciprocal Recommendation Systems (RRS) [6] to help the two stakeholders achieve mutual benefits, which is also far from a trivial task.

To tackle the above problems, in this paper, we propose a novel **Bilateral Multi-Behavior mODEling (BAMBOO)** method for reciprocal recommendation in online recruitment, which models the interactions between job seekers and recruiters from two different perspectives, respectively *expectation perspective* and *competitiveness perspective*. Specifically, for the expectation perspective, we propose to format the historical behaviors of users as *bilateral multi-behavior sequences*, and we design a transformer-based sequence modeling architecture with behavior type embedding to get the representations of these sequences. Then, for the competitiveness perspective, we propose to build a *bilateral heterogeneous interaction graph* which can distinguish the active and passive behaviors by designing different types of edges, and we design a transformer-based heterogeneous graph neural network to get the node representations in this graph. To further enhance the representation learning abilities of these two models, we design a contrastive learning loss between the output embedding from the two different perspectives. By considering

both expectation and competitiveness perspectives, we can fully and efficiently model the multi-typed user behaviors in recruitment platforms, leading to better representations of job seekers and recruiters. Moreover, in order to carry out the reciprocal recommendation between job seekers and recruiters, we propose to decompose the matching probability into the product of two different probabilities, respectively the application probability of the active part and the acceptance probability of the passive part. We use different output models for the prediction of different probabilities, and train these models following the multi-task learning paradigm. Finally, we conduct extensive experiments to demonstrate the effectiveness of our model. Both online A/B test and offline experiments on real-world datasets clearly validate the effectiveness of our BAMBOO model compared with other state-of-the-art baseline methods. To the best of our knowledge, the contribution of this article can be summarized as follows:

- We develop a novel end-to-end framework named BAMBOO to model the multi-typed user behaviors in online recruitment platforms from two different perspectives with two transformer-based models.
- We propose to enhance the learning abilities of the two models by contrastive learning. Furthermore, we propose to train the reciprocal recommendation model following the multi-task learning paradigm.
- We conduct extensive experiments on real-world datasets and online A/B test, which validate the effectiveness of our BAMBOO framework.

II. RELATED WORK

In this section, we will summarize the related works in the following three categories, respectively RS for online recruitment, multi-behavior recommendation, and heterogeneous graph neural networks.

A. RS for Online Recruitment

With the rapid development of online recruitment services, a variety of emerging recommendation methods, especially reciprocal recommendation methods, have been witnessed for benefiting both job seekers and recruiters. The methods can be broadly classified into two categories, respectively text-based methods [2], [3], [5], [7] and behavior-based methods [4], [8], [9]. For the text-based methods, Zhu et al. [2] first formulated this task as a joint representation learning problem, and utilized CNN-based end-to-end models to get the representation of job seekers and recruiters from resumes and job descriptions. Along this line, Qin et al. [3] enhanced the above model by taking the abilities into consideration, and used RNN-based models and attention mechanisms for hierarchical ability-aware representation. Le et al. [5] proposed an interpretable model to match job seekers and recruiters from different intentions in a multi-task learning paradigm. For the behavior-based methods, Yang et al. [4] proposed to build an interaction graph between job seekers and recruiters, and designed a dual-perspective message passing neural network to model the directed interactions between the two stakeholders. Fu et al. [8] proposed to utilize memory networks to get the representation of the multi-behavior

sequences of different job seekers and recruiters. Recently, some studies also focus on leveraging Large Language Models (LLM) for job recommendation [10], [11], [12]. However, the above studies only focus on getting the representation of user behaviors from a single perspective, which limits them to achieve satisfactory performance. Different from these studies, we propose to model the user behaviors from two different perspectives.

B. Multi-Behavior Recommendation

Most of the traditional recommendation systems are designed for single user behavior type, e.g., *click* or *rating* [13], [14]. However, in platforms like online retailing platforms, users can *view* products, *tag* them as favorites, and finally *purchase* them. Recently, several studies on modeling the behavior heterogeneity have emerged [15], [16], [17], [18], [19], [20], [21], [22], [23]. For example, Gao et al. [15] proposed a neural multi-task recommendation solution, which can enhance the performance of the model by capturing the ordinal relations among different behavior types. Jin et al. [16] proposed to construct a unified heterogeneous graph based on multiple types of behavioral data, and designed a multi-behavior graph convolutional network, which can learn behavior strength by user-item propagation layer and capture behavior semantics by item-item propagation layer. Xia et al. [17] further proposed a knowledge-enhanced hierarchical graph transformer, which can explicitly discriminate the importance of different user-item interaction types. Recently, many studies on extending multi-behavior modeling approaches for sequential data have emerged. For example, Xia et al. [18] presented a temporal graph transformer model to capture the dynamic cross-type behavior inter-dependency. Yang et al. [19] proposed a multi-behavior hypergraph learning paradigm to distill the item-specific multi-behavior correlations with global and customized sequential context injection. Different from the above studies, the multi-behavior sequences in online recruitment platforms are generated by two different stakeholders, which brings us new challenges.

C. Heterogeneous Graph Neural Networks

Recently, many studies on extending deep learning approaches for graph data, i.e., Graph Neural Networks (GNN), have emerged [24], [25], [26], [27], [28]. These methods mainly designed for graphs with only single type of node and edge. However, many graph structures have multiple types of nodes and edges, i.e., heterogeneous graphs. To solve this problem, many studies on heterogeneous graph neural networks have emerged [29], [30], [31], [32], [33], [34]. For example, Schlichtkrull et al. [29] introduced relational graph convolutional networks to model knowledge graphs with multiple types of edges. Wang et al. [30] proposed a heterogeneous graph neural network based on hierarchical attention, including node-level and semantic-level attentions, to take the importance of nodes and meta-paths into consideration simultaneously. Hu et al. [31] further presented the heterogeneous graph transformer architecture for modeling Web-scale heterogeneous graphs, with

node- and edge-type dependent parameters to characterize the heterogeneous attention over each edge. Inspired by the above studies, in this paper, we utilize heterogeneous graph neural networks to describe the interactions between job seekers and recruiters from a competitiveness perspective.

III. PRELIMINARIES

In this section, we first introduce the real-world datasets used in this paper. Then, we introduce how to construct the bilateral multi-behavior sequences and the bilateral heterogeneous interaction graph. Finally, we propose the problem formulation of reciprocal recommendation.

A. Data Description and Preprocessing

The real-world datasets used in this paper comes from one of the largest online recruitment platform in China, ranging from Mar. 1, 2023, to Mar. 22, 2023. In our datasets, each job seeker and recruiter is de-linked from the production system by securely hashing with one-time salt mapping. In this platform, job seekers have their corresponding profile features, e.g., age, major, educational background, etc., and the recruiters also have profile features, e.g., company scale, city, position, salary, etc. We collected all the interaction behaviors between job seekers and recruiters, which are naturally multi-typed. Indeed, the two stakeholders need to go through many stages before reaching an agreement. For example, as shown in Fig. 1, a job seeker first needs to view the recommendation list with multiple job descriptions, then clicks on the specific one he/she likes, and sends a message to chat with the recruiter. Then, the recruiter needs to pay attention to his/her message and talks back, and they can finally reach an agreement. Note that the above process is an example of a job seeker being the active party, while the situation is completely symmetrical when the recruiter is the active party. To summarize this problem and capture the essence of reciprocal recommendation, we propose to use a *direction* label and a *status* label to describe the above process. Specifically, the *direction* label is used to indicate who is the initiator of a specific interaction, i.e., who is the *active* party. We set the value of the *direction* label as:

$$direction = \begin{cases} 0 & \text{if a job seeker is the active party.} \\ 1 & \text{if a recruiter is the active party.} \end{cases} \quad (1)$$

The *status* label is used to indicate the final result of a specific interaction. In this paper, we summarize the interaction result into the following three categories: 1) *Neglect*. The active party is not attracted to the passive party. For example, a job seeker viewed a job description but did not send a message to the recruiter. 2) *Apply*. The active party is attracted to the passive party, but the passive party is not attracted to the active party. For example, a job seeker sent a message to a recruiter, but the recruiter did not talk back. 3) *Match*. The active and passive parties attracted each other and reached an agreement. We set

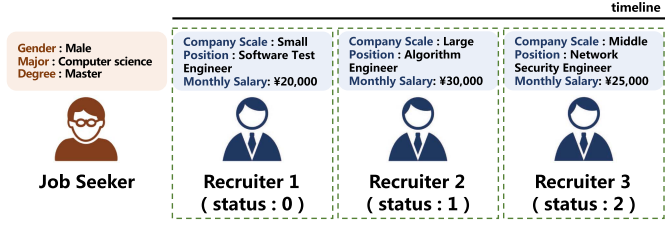


Fig. 2. An example of the bilateral multi-behavior sequence of a specific job seeker.

the value of the status label as:

$$status = \begin{cases} 0 & \text{if active party neglected passive party.} \\ 1 & \text{if active party applied to passive party.} \\ 2 & \text{if active party matched with passive party.} \end{cases} \quad (2)$$

Note that our method has good scalability, i.e., we can model more types of interactions by simply increasing the number of status labels. Furthermore, we also collected the context features corresponding to each interaction record, e.g., the hour of the day. Based on the above, we propose the final format of each piece of data used in this paper as $[job\ seeker\ id, recruiter\ id, job\ seeker\ features, recruiter\ features, context\ features, direction, status, timestamp]$. Note that we filled the missing value in features with padding characters. Furthermore, since the candidates that can attract the interest of a specific user are only a small part of the total candidates, we conducted negative sampling on the data with status 0.

B. Multi-View Data Construction

To support the multi-view modeling in this paper, we propose the following methods to construct the bilateral multi-behavior sequences and the bilateral heterogeneous interaction graph. Specifically, we first split the datasets into four parts, respectively *history data*, *training data*, *validation data*, and *test data*. We further split the *history data* into two parts, respectively *job seeker history data* with direction label 0 and *recruiter history data* with direction label 1. In order to construct the bilateral multi-behavior sequences, for each job seeker/recruiter, we collected the corresponding data from the job seeker/recruiter history data with different status labels, and sorted the data in chronological order as shown in Fig. 2. Then, as shown in the left part of Fig. 3, we first build the initial interaction graph between job seekers and recruiters based on the historical data. However, this approach fails to capture the essence of bidirectional matching. Therefore, as shown in the right part of Fig. 3, we propose to construct the bilateral heterogeneous interaction graph, which can distinguish the active/passive behaviors and attracted/unattracted results with four types of edges in a heterogeneous graph. We will demonstrate the superiority of this way to build the interaction graph in the following experiments part.

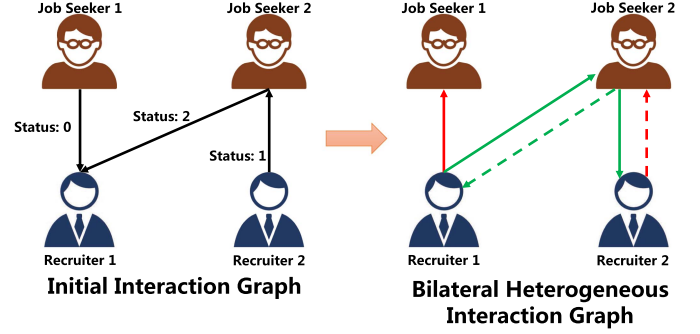


Fig. 3. An illustration of the bilateral heterogeneous interaction graph. We convert the edges in the initial graph to bilateral edges based on activeness and attractiveness. In this way, we get four different types of edges. We use green color for *attracted* and red color for *unattracted*, and we use solid lines for *active* and dotted lines for *passive*. Note that in order to facilitate the message passing process in graph neural networks, we reversed the direction of edges.

TABLE I
MATHEMATICAL NOTATIONS

Symbol	Description
Θ	Model parameters
$\mathcal{G}$	The bilateral heterogeneous interaction graph
$\mathcal{H}^s$	The bilateral multi-behavior sequence of job seeker s
$\mathcal{H}^r$	The bilateral multi-behavior sequence of recruiter r
$\mathbf{s}$	The profile features of job seeker s
$\mathbf{r}$	The profile features of recruiter r
$\mathbf{c}$	The context features
$\mathbf{n}_t^l$	The node embedding of node t in the l -th GNN layer.
$p_{apply}^{a \rightarrow b}$	The probability that a sends application to b .
$p_{acc}^{a \rightarrow b}$	The probability that b accept the application from a .
$p_{match}^{a \rightarrow b}$	The probability that a matches with b .
ϕ	The node type mapping function
ψ	The edge type mapping function
$\begin{bmatrix} \cdot \\ \cdot \end{bmatrix}$	The stack of two vectors/matrices
$\begin{bmatrix} \cdot \\ \cdot \end{bmatrix}$	The concatenation of two vectors/matrices
$MLP(\cdot)$	Multilayer Perceptron with ReLU Activation Function

C. Problem Formulation

Here we introduce the problem formulation of the reciprocal recommendation problem in online recruitment. For facilitating illustration, Table I lists some mathematical notations used throughout this paper. Based on the above preprocessing method, given a job seeker $s \in \mathcal{S}$ and a recruiter $r \in \mathcal{R}$, we can first format the profile features of s as $\mathbf{s} = [p_1^s, \dots, p_m^s]$ and the profile features of r as $\mathbf{r} = [p_1^r, \dots, p_n^r]$, where p_i^s and p_i^r are the i -th feature of s and r . We can also format a context feature vector $\mathbf{c}$ as $\mathbf{c} = [p_1^c, \dots, p_t^c]$, where p_i^c is the i -th context feature. Then, we format the historical bilateral multi-behavior sequence of s and r as $\mathcal{H}^s = \{\mathcal{I}_1^s, \dots, \mathcal{I}_j^s\}$ and $\mathcal{H}^r = \{\mathcal{I}_1^r, \dots, \mathcal{I}_k^r\}$, where $\mathcal{I}_i^s = \{\mathbf{r}_i^s, l_i^s\}$ and $\mathcal{I}_i^r = \{\mathbf{s}_i^r, l_i^r\}$. $\mathbf{r}_i^s$ is the profile feature of the i -th recruiter in the history record of job seeker s and l_i^s is the corresponding status label. $\mathbf{s}_i^r$ is the profile feature of the i -th job seeker in the history record of recruiter r and l_i^r is the corresponding status label. Furthermore, we format the bilateral heterogeneous interaction graph as $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$, associated with a node type mapping function $\phi: \mathcal{V} \rightarrow \mathcal{A}$ and an edge type mapping function $\psi: \mathcal{E} \rightarrow \mathcal{R}$, where $\mathcal{A}$ contains two different types of users and $\mathcal{R}$ contains four different types of edges.

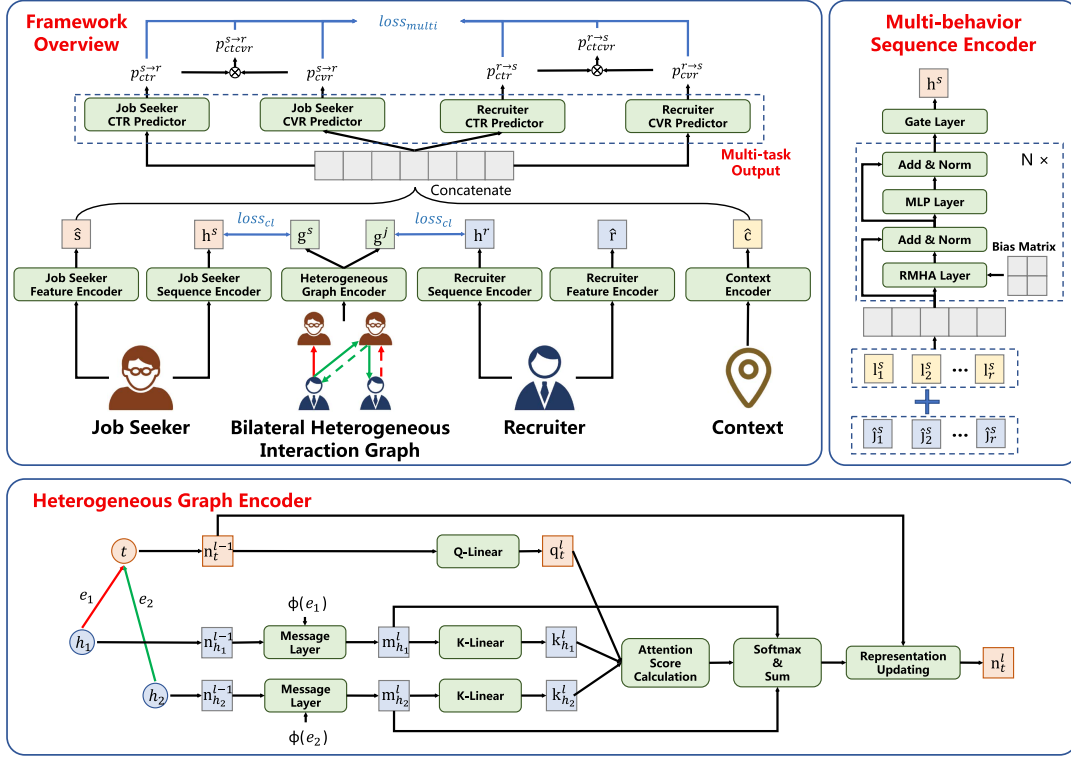


Fig. 4. An overview of our BAMBOO model.

Along this line, the problem of reciprocal recommendation problem in online recruitment can be formulated as follows:

Definition 1 (Reciprocal Recommendation): Given a job seeker s with the corresponding profile features $\mathbf{s}$ and the historical sequence $\mathcal{H}^s$, a recruiter r with the corresponding profile features $\mathbf{r}$ and the historical sequence $\mathcal{H}^r$, the context features $\mathbf{c}$ corresponding to the interaction, the heterogeneous graph $\mathcal{G}$, and the direction label, the goal of reciprocal recommendation is to estimate the probability that the two stakeholders can finally reach an agreement.

IV. TECHNICAL DETAILS

In this section, we will introduce the framework of our model in detail. As shown in Fig. 4, our model mainly consists of four components, respectively *feature encoder*, *multi-behavior sequence encoder*, *heterogeneous graph encoder*, and *multi-task output layer*.

A. Feature Encoder

Given a job seeker feature vector $\mathbf{s}$, a recruiter feature vector $\mathbf{r}$, and a context vector $\mathbf{c}$, the first thing we need to do is to get their representations. In this paper, we use a representation method based on mean-pooling to construct the feature encoders for these features vectors as follows:

$$\hat{\mathbf{s}} = \frac{1}{\|\mathbf{s}\|_1} \mathbf{s} \mathbf{E}^s, \quad \hat{\mathbf{r}} = \frac{1}{\|\mathbf{r}\|_1} \mathbf{r} \mathbf{E}^r, \quad \hat{\mathbf{c}} = \frac{1}{\|\mathbf{c}\|_1} \mathbf{c} \mathbf{E}^c, \quad (3)$$

where $\mathbf{E}^s \in \mathbb{R}^{m \times d}$, $\mathbf{E}^r \in \mathbb{R}^{n \times d}$, $\mathbf{E}^c \in \mathbb{R}^{t \times d}$ are the embedding matrices, d denotes the embedding dimension, and $\|\cdot\|_1$ denotes the 1-norm of a vector. Using Fig. 2 as an example, we can get the feature embedding of job seeker and recruiter attributes such as Major, Degree, and Company Scale. The final representation vector will be obtained using the mean pooling method.

B. Multi-Behavior Sequence Encoder

As shown in Fig. 2, we collected the historical behaviors of job seekers and recruiters, and constructed bilateral multi-behavior sequences for them. For utilizing these bilateral multi-behavior sequences to get the representation from the expectation perspective, we propose the following transformer-based multi-behavior sequence encoder. In this section, we will introduce the details of this model.

Specifically, given a job seeker s with the historical sequence $\mathcal{H}^s = \{\mathcal{I}_1^s, \dots, \mathcal{I}_T^s\}$ where $\mathcal{I}_i^s = \{\mathbf{r}_i^s, \mathbf{l}_i^s\}$, we first input the feature vector $\mathbf{r}_i^s$ of the i -th recruiter into the feature encoder we propose in Section IV-A, and get the corresponding embedding $\hat{\mathbf{r}}_i^s$. By this way, we can format the embedding of all the recruiters in $\mathcal{H}^s$ as a matrix $\hat{\mathbf{R}}^s = [\hat{\mathbf{r}}_1^s, \dots, \hat{\mathbf{r}}_T^s]$.

After that, we need to capture the influence of different bilateral behaviors, which is the main challenge of modeling user behaviors in online recruitment. For example, a job seeker may want to become an algorithm engineer and send messages to some recruiters. If most of the recruiters recognize his abilities and send offers to him, then he may be encouraged and continue to find relevant jobs. Otherwise, if most recruiters reject his

application, then he may realize that he is not suitable for this position and try to apply for other positions. Along this line, we first convert each label l , e.g., the status in Fig. 2, to a representation vector $\mathbf{l}$ by an embedding matrix and get the label embedding matrix $\mathbf{L}^s = [\mathbf{l}_1^s, \dots, \mathbf{l}_j^s]$ corresponding to $\mathcal{H}^s$, where $\mathbf{l}_i^s$ is the corresponding embedding to l_i^s . Then, we fuse the label information to the behavior sequence by summing up the above two matrices as:

$$\tilde{\mathbf{R}}^s = \hat{\mathbf{R}}^s + \mathbf{L}^s. \quad (4)$$

After that, we propose to leverage a transformer-based method to get the representation of $\tilde{\mathbf{R}}^s$. Specifically, given three input matrices $\mathbf{Q} \in \mathbb{R}^{l_q \times d}$, $\mathbf{K} \in \mathbb{R}^{l_v \times d}$ and $\mathbf{V} \in \mathbb{R}^{l_v \times d}$, the attention operation in vanilla transformer architecture [35] can be format as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}} \right) \mathbf{V}. \quad (5)$$

Furthermore, in order to capture the sequential information, [35] proposed a position encoding method to enhance the above attention calculation process. Recently, Liu et al. [36] further proposed that adjusting the attention scores based on the relative positions can achieve better results. Therefore, in this paper, we define the relative position-aware attention as:

$$\text{R-Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}} + \mathbf{B} \right) \mathbf{V}, \quad (6)$$

where $\mathbf{B} \in \mathbb{R}^{l_q \times l_v}$ is the relative position scores to learn, $\mathbf{B}_{i,j}$ is the bias score to fix the attention score between items are position i and position j . The multi-head attention calculation can be formatted as:

$$\begin{aligned} \text{RMHA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) &= [\text{head}_1; \dots; \text{head}_{l_h}] \mathbf{W}^I, \\ \text{head}_i &= \text{R-Attention}(\mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}\mathbf{W}_i^K, \mathbf{V}\mathbf{W}_i^V), \end{aligned} \quad (7)$$

where $\mathbf{W}_i^Q, \mathbf{W}_i^K, \mathbf{W}_i^V \in \mathbb{R}^{d \times d/l_h}$ and $\mathbf{W}^I \in \mathbb{R}^{d \times d}$ are the parameters to learn. l_h is the number of heads. Based on the above, the transformer layer for $\tilde{\mathbf{R}}^s$ can be defined as:

$$\begin{aligned} \mathbf{H}^s &= \text{LayerNorm}(\mathbf{M}^s + \text{MLP}(\mathbf{M}^s)), \\ \mathbf{M}^s &= \text{LayerNorm}(\tilde{\mathbf{R}}^s + \text{RMHA}(\tilde{\mathbf{R}}^s, \tilde{\mathbf{R}}^s, \tilde{\mathbf{R}}^s)), \end{aligned} \quad (8)$$

where $\text{LayerNorm}(\cdot)$ is proposed in [37]. Finally, we convert the embedding matrix $\mathbf{H}^s$ to a vector $\mathbf{h}^s$ by the following gating layer:

$$\mathbf{h}^s = (\text{softmax}(\text{ReLU}(\mathbf{H}^s \mathbf{W}_a + \mathbf{v}_a) \mathbf{W}_g))^\top \mathbf{H}^s, \quad (9)$$

where $\mathbf{W}_a \in \mathbb{R}^{d \times d}$, $\mathbf{v}_a \in \mathbb{R}^{1 \times d}$ and $\mathbf{W}_g \in \mathbb{R}^{d \times 1}$ are all trainable parameters. In this way, given a job seeker s and a recruiter r , we can get the embedding of their historical bilateral multi-behavior sequence $\mathbf{h}^s$ and $\mathbf{h}^r$ symmetrically.

C. Heterogeneous Graph Encoder

Based on the sequence encoder we propose in Section IV-B, we get the user embedding from the expectation perspective, which contains information about the evolution process of user interest and expectation. To further solve the challenges brought

by the *diversity* and *bidirectionality* characteristics, and capture the market competition between recruiters and job seekers we propose to construct the bilateral heterogeneous interaction graph in Section III-A as shown in Fig. 3. In this section, we will introduce the encoding method we used for the heterogeneous graph to get the user embedding from the competitiveness perspective.

Specifically, given a directed edge $\langle h, e, t \rangle$, where h is the head node, e is the edge, and t is the tail node, we can first get the node types $\phi(s)$, $\phi(t)$ and edge type $\psi(e)$. Then, we calculate the message vector passed from h to t in the l -th layer of the heterogeneous graph neural network as:

$$\text{MSG}^l(h, e, t) = (\mathbf{n}_h^{l-1} \mathbf{W}_{\psi(e)}^l + \mathbf{b}_{\psi(e)}^l) \mathbf{V}_{\psi(e)}^l, \quad (10)$$

where $\mathbf{n}_h^{l-1}$ is the node representation of h in the $(l-1)$ -th layer. $\mathbf{W}_{\psi(e)}^l \in \mathbb{R}^{d \times d}$, $\mathbf{V}_{\psi(e)}^l \in \mathbb{R}^{d \times d}$, and $\mathbf{b}_{\psi(e)}^l \in \mathbb{R}^{1 \times d}$ are all trainable parameters corresponding to edge type $\psi(e)$. Then, inspired by the Heterogeneous Graph Transformer (HGT) model [31], we propose to leverage the attention mechanism to aggregate the message from all the neighbors of h as follows:

$$\begin{aligned} \text{ATT}^l(h, e, t) &= \text{Softmax}_{\forall h \in N(t)} (\text{ATT-Score}^l(h, e, t)), \\ \text{ATT-Score}^l(h, e, t) &= (\mathbf{k}_h^l \cdot \mathbf{q}_t^l) \cdot \frac{\mu \langle \phi(h), \psi(e), \phi(t) \rangle}{\sqrt{d}}, \\ \mathbf{k}_h^l &= \text{K-Linear}_{\phi(h)}(\mathbf{r}_h^{l-1}), \\ \mathbf{q}_t^l &= \text{Q-Linear}_{\phi(t)}(\mathbf{r}_t^{l-1}), \end{aligned} \quad (11)$$

where $\text{K-Linear}_{\phi(h)}(\cdot)$ and $\text{Q-Linear}_{\phi(t)}(\cdot)$ are the linear layers to project the node representation from the $(l-1)$ -th layer to the key and query vectors in the l -th layer. $\mathbf{k}_h^l \cdot \mathbf{q}_t^l$ is the inner product of these two vectors. $\mu \in \mathbb{R}^{|\mathcal{A}| \times |\mathcal{R}| \times |\mathcal{A}|}$ are trainable parameters for the general significance of each meta relation type. $N(t)$ are the neighbors of t . d is the dimension of vectors. Based on the above message passing result, we can finally define the node representation updating process as:

$$\hat{\mathbf{n}}_t^l = \sum_{\forall h \in N(t)} (\text{ATT}^l(h, e, t) \cdot \text{MSG}^l(h, e, t)), \quad (12)$$

$$\mathbf{n}_t^l = \text{Linear}_{\phi(t)}(\hat{\mathbf{n}}_t^l) + \mathbf{n}_t^{l-1}, \quad (13)$$

where $\text{Linear}_{\phi(t)}(\cdot)$ is the linear projection layer.

Based on the above message passing process, given a job seeker s and a recruiter r , we can get the embedding of their corresponding nodes in the bilateral heterogeneous interaction graph as $\mathbf{g}^s$ and $\mathbf{g}^r$.

D. Multi-Task Output Layer

Now we can calculate the match probabilities between job seekers and recruiters according to their representations. Based on the above sections, given a job seeker s and a recruiter r with the interaction context, we can get the feature embedding as $\hat{\mathbf{s}}$, $\hat{\mathbf{r}}$, and $\hat{\mathbf{c}}$, the sequence embedding as $\mathbf{h}^s$ and $\mathbf{h}^r$, and the embedding from the interaction graph as $\mathbf{g}^s$ and $\mathbf{g}^r$. A straightforward

way to solve the prediction task is to directly input the above vectors into a prediction model, such as an MLP layer, to get the output prediction. However, this naive method ignores the following important features of reciprocal recommendation in online recruitment: 1) There is an active party and a passive party in an event; 2) The final matching can be reached only when the two parties are mutually satisfied; 3) The active and passive preferences of the same person may be different. For example, a person may only actively send requests to highly satisfied candidates, but may passively accept applications from relatively less satisfied candidates. Therefore, in this section, we propose to design a multi-task output layer to solve the above challenges.

Specifically, suppose the job seeker s is the active party and the recruiter r is the passive party. To estimate the match probability of the two stakeholders, we propose to first estimate the probability of the active party initiating a chat with the passive party. We use an MLP layer for the calculation as:

$$p_{apply}^{s \rightarrow r} = \sigma(\text{MLP}_{apply}^{s \rightarrow r}([\hat{s}; \hat{r}; \hat{c}; \mathbf{h}^s; \mathbf{h}^r; \mathbf{g}^s; \mathbf{g}^r])), \quad (14)$$

where $\sigma(\cdot)$ is the sigmoid function. The superscript $s \rightarrow r$ means that the job seeker s is the active party and the recruiter r is the passive party. Then, we propose to estimate the probability that the passive party accepts the request from the active party. Another MLP layer is used for the calculation as:

$$p_{acc}^{s \rightarrow r} = \sigma(\text{MLP}_{acc}^{s \rightarrow r}([\hat{s}; \hat{r}; \hat{c}; \mathbf{h}^s; \mathbf{h}^r; \mathbf{g}^s; \mathbf{g}^r])). \quad (15)$$

Finally, we propose to decompose the matching probability into the product of the above two probabilities as:

$$p_{match}^{s \rightarrow r} = p_{apply}^{s \rightarrow r} * p_{acc}^{s \rightarrow r}. \quad (16)$$

Symmetrically, if recruiter r is the active party, we have:

$$p_{apply}^{r \rightarrow s} = \sigma(\text{MLP}_{apply}^{r \rightarrow s}([\hat{s}; \hat{r}; \hat{c}; \mathbf{h}^s; \mathbf{h}^r; \mathbf{g}^s; \mathbf{g}^r])), \quad (17)$$

$$p_{acc}^{r \rightarrow s} = \sigma(\text{MLP}_{acc}^{r \rightarrow s}([\hat{s}; \hat{r}; \hat{c}; \mathbf{h}^s; \mathbf{h}^r; \mathbf{g}^s; \mathbf{g}^r])), \quad (18)$$

and the matching probability can be written as:

$$p_{match}^{r \rightarrow s} = p_{apply}^{r \rightarrow s} * p_{acc}^{r \rightarrow s}. \quad (19)$$

E. Training Strategies

In this section, we will introduce the training strategies of our BAMBOO model, including a multi-task learning loss function and a contrastive learning loss function.

1) *Multi-Task Learning*: For the multiple output probabilities from the multi-task output layer, we propose to treat the p_{apply} , p_{acc} , and p_{match} prediction as three different tasks with internal relevance. We adopt the BCE loss as the loss function for each task, and the difference between these tasks is the setting of ground truth labels. Specifically, suppose the job seekers are active, we have the following loss function for application prediction as:

$$\begin{aligned} \mathcal{L}_1 = & -\frac{1}{N_1} \sum_{s \rightarrow r} [l_{apply}^{s \rightarrow r} \log(p_{apply}^{s \rightarrow r}) \\ & + (1 - l_{apply}^{s \rightarrow r}) \log(1 - p_{apply}^{s \rightarrow r})], \end{aligned} \quad (20)$$

where N_1 is the total number of data, and $l_{apply}^{s \rightarrow r}$ is the ground truth label for the above application prediction task which has the following value:

$$l_{apply}^{s \rightarrow r} = \begin{cases} 0 & \text{if } status = 0. \\ 1 & \text{if } status = 1 \text{ or } status = 2. \end{cases} \quad (21)$$

Then, we have the following loss function for acceptance prediction as:

$$\begin{aligned} \mathcal{L}_2 = & -\frac{1}{N_2} \sum_{\substack{s \rightarrow r \\ status \neq 0}} [l_{acc}^{s \rightarrow r} \log(p_{acc}^{s \rightarrow r}) \\ & + (1 - l_{acc}^{s \rightarrow r}) \log(1 - p_{acc}^{s \rightarrow r})], \end{aligned} \quad (22)$$

where N_2 is the total number of data, and $l_{acc}^{s \rightarrow r}$ is the ground truth label for the above CVR prediction task which has the following value:

$$l_{acc}^{s \rightarrow r} = \begin{cases} 0 & \text{if } status = 1. \\ 1 & \text{if } status = 2. \end{cases} \quad (23)$$

Note that we exclude the data with $status = 0$, since the calculation of the acceptance probability is meaningless if the active party did not send applications to the passive party. Finally, we have the following loss function for matching prediction as:

$$\begin{aligned} \mathcal{L}_3 = & -\frac{1}{N_3} \sum_{s \rightarrow r} [l_{match}^{s \rightarrow r} \log(p_{match}^{s \rightarrow r}) \\ & + (1 - l_{match}^{s \rightarrow r}) \log(1 - p_{match}^{s \rightarrow r})], \end{aligned} \quad (24)$$

where N_3 is the total number of data, and $l_{match}^{s \rightarrow r}$ is the ground truth label for the above matching prediction task which has the following value:

$$l_{match}^{s \rightarrow r} = \begin{cases} 0 & \text{if } status = 0 \text{ or } status = 1. \\ 1 & \text{if } status = 2. \end{cases} \quad (25)$$

Symmetrically, if the recruiter is the active party, we can also get three corresponding loss, respectively $\mathcal{L}_4$, $\mathcal{L}_5$, and $\mathcal{L}_6$. Based on the above, the multi-task learning loss can be written as follows:

$$\mathcal{L}_{multi} = \mathcal{L}_1 + \mathcal{L}_2 + \mathcal{L}_3 + \mathcal{L}_4 + \mathcal{L}_5 + \mathcal{L}_6. \quad (26)$$

Note that all the probabilities in this section are computed based on the vectors in Sections IV-A, IV-B, and IV-C. Therefore, through the above loss function, all the parameters in the models discussed in Sections IV-A, IV-B, and IV-C will be updated simultaneously during the training process.

2) *Contrastive Learning*: We propose to get the user embedding from the expectation perspective in Section IV-B and from the competitiveness perspective in Section IV-C. Furthermore, inspired by the recent studies on contrastive learning [38], [39], we proposed that these two parts can actually learn from each other and improve the overall performance of the final prediction tasks. Therefore, we propose to design contrastive loss between the output from the two different models. Specifically, given a specific user, we hope the sequence embedding of this user can be as similar to his own graph embedding as possible, compared to the graph embedding of other users. Along this line, the

TABLE II
STATISTICS OF THE DATASETS

description	arts	manufacturing	service	technology
The number of records	4,015,530	5,050,659	20,806,849	12,283,147
The number of records in job seeker history data	1,765,816	1,579,943	4,096,324	5,871,771
The number of records in recruiter history data	2,014,830	3,234,667	15,873,339	5,287,412
The number of records in training data	85,487	94,748	351,503	481,252
The number of records in validation data	69,559	68,621	255,218	343,816
The number of records in testing data	79,838	72,680	230,465	298,896
The number of records with <i>status</i> = 0	1,418,483	1,433,620	6,253,095	4,854,061
The number of records with <i>status</i> = 1	2,028,202	2,875,602	12,899,090	5,855,626
The number of records with <i>status</i> = 2	568,845	741,437	1,654,664	1,573,460
The number of records with <i>direction</i> = 0	1,870,857	1,679,995	4,343,715	6,418,448
The number of records with <i>direction</i> = 1	2,144,673	3,370,664	16,463,134	5,864,699
The number of job seekers	833,798	1,602,136	5,105,633	1,199,317
The number of recruiters	369,121	475,736	1,224,287	831,060
The number of edges in the bilateral interaction graph	792,765	704,152	2,676,168	2,976,078

contrastive loss used in this paper can be formatted as follows:

$$\mathcal{L}_7 = -\frac{1}{N_s} \sum_{s \in \mathcal{S}} \log \frac{\exp(\text{sim}(\mathbf{h}^s, \mathbf{g}^s))}{\sum_{\hat{s} \in \mathcal{S}} \exp(\text{sim}(\mathbf{h}^s, \mathbf{g}^{\hat{s}}))}, \quad (27)$$

$$\mathcal{L}_8 = -\frac{1}{N_r} \sum_{r \in \mathcal{R}} \log \frac{\exp(\text{sim}(\mathbf{h}^r, \mathbf{g}^r))}{\sum_{\hat{r} \in \mathcal{R}} \exp(\text{sim}(\mathbf{h}^r, \mathbf{g}^{\hat{r}}))}, \quad (28)$$

$$\mathcal{L}_{cl} = \mathcal{L}_7 + \mathcal{L}_8, \quad (29)$$

where N_s and N_r are the total numbers of job seekers and recruiters. $\text{sim}(\cdot, \cdot)$ is the similarity evaluation function for two different vectors. In practice, we use the inner product of two vectors as the function $\text{sim}(\cdot, \cdot)$. Furthermore, we use negative sampling to select different users in order to reduce the calculation cost.

Based on the above, the final training loss function for our BAMBOO model can be formatted as:

$$\mathcal{L}_{total} = \alpha * \mathcal{L}_{multi} + (1 - \alpha) * \mathcal{L}_{cl} + \lambda * \|\Theta\|_2^2, \quad (30)$$

where α is a hyper-parameter, which is used to control the trade-off between $\mathcal{L}_{multi}$ and $\mathcal{L}_{cl}$, and Θ is the parameter set of our model. L_2 regularization is applied to prevent the overfitting of our model.

V. EXPERIMENTS

In this section, we evaluate the proposed model with a number of competitive baselines. Meanwhile, we will further present the discussions and case studies on the reciprocal recommendation system for online recruitment. The experiments are mainly designed to answer the research questions as follows:

- *RQ1*: How does our model perform compared with state-of-the-arts recommendation methods?
- *RQ2*: Whether the specially designed parts of our framework effective and how do different settings influence the effectiveness of our model?
- *RQ3*: Whether our study can reveal some interesting findings for online recruitment?

A. Data Statistics

As we mentioned in Section III-A, the real-world datasets come from one of the largest online recruitment platform in China. We collected four different datasets from four different domains, respectively arts, manufacturing (manu), service (serv), and technology (tech). Detailed statistics of these four datasets are shown in Table II. From this table, we can draw some interesting conclusions. For example, in the service domain, the number of records with *direction* = 1 is significantly more than the number of records with *direction* = 0, which indicates that the job seekers in the service domain tend to accept invitations passively. On the contrary, job seekers in the technology domain are more inclined to actively seek suitable jobs on their own.

B. Baselines Methods

To evaluate the performance of our models, we selected several state-of-art methods as baselines. Specifically, we first chose two popular recommendation approaches as follows:

- *DIN* [40]: This model utilizes attention mechanisms to capture the user interest from the clicked items. Both feature embedding and user interest embedding are used as the input for the multi-task output layer.
- *DIEN* [41]: This model further adds a sequential modeling part to capture the evolution of user interest compared with the DIN model.

Then, we chose two popular heterogeneous graph neural network approaches as follows:

- *HAN* [30]: This model is a popular embedding method for heterogeneous graphs. Feature embedding and graph node embedding are used as the input for the multi-task output layer.
- *HGT* [31]: This model is a state-of-the-art embedding method for heterogeneous graphs.

After that, we chose three different multi-behavior modeling methods as follows:

- *MBGCN* [16]: This model takes use of multiple types of interactions by graph convolutional network. Feature embedding and graph node embedding are used as the input for the multi-task output layer.

- *MBDIT*: In this model, we add behavior label embedding to the DIT model [42].
- *MBGRU*: In this model, we add behavior label embedding to the GRU4Rec model [43].

Note that all the above models are not specially designed for reciprocal recommendation. Therefore, our multi-task output layer is utilized for the above methods. Finally, we chose three person-job fit models as follows:

- *PJFNN* [2]: This model is a person-job fit model based on text matching. In this paper, we utilize the feature embedding of job seekers, recruiters, and context as the input for the multi-task output layer.
- *IPJF* []: This model utilizes Mixture-of-Experts BIPar-MoEBIPar module to independently learn the preference representations of job seekers and recruiters.
- *DPGNN* [4]: This model is a state-of-the-art two-way selection person-job fit model using graph neural network model and contrastive learning.
- *BOSS* [44]: This model utilizes Mixture-of-Experts (MoE) module to independently learn the preference representations of job seekers and recruiters.

C. Evaluation

To evaluate the performance of different models, we selected *AUC* and *LogLoss (LL)* as the evaluation metrics. Specifically, we evaluated the performance of different models on matching prediction in both the job seeker side and the recruiter side. It is worth noting that in the definition in Section III-C, we focus on improving the prediction of the final matching probability, which is the ultimate goal of reciprocal recommendation. However, our model is a multi-task model, and in real-world scenarios, evaluating the probability of active application is also meaningful. Therefore, in the experiment part, we will simultaneously evaluate the effect of the model on both matching probability estimation and application probability estimation. Specifically, we use p_{match} as the recommendation probability for matching probability estimation, and p_{apply} as the recommendation probability for application probability estimation. Note that matching probability estimation is still our primary task, while application probability estimation is auxiliary.

D. Implementation Details

We implemented our model using a GeForce RTX 3090 GPU with 24 GB memory on a Linux machine. We set the dimension of all the feature embedding, sequence embedding, and graph node embedding as 64. In the process of model training, we used the Adam optimizer [45] for parameter optimization. We set the learning rate as 0.001 for training, and we set the mini-batch size as 1024. The parameters of baselines were set up similarly to our method and were all tuned to be optimal to ensure fair comparisons. We report the performance on the test set for the model that performed the best on the validation set, and we run each model 10 times with different random seeds for the t-test.

E. Overall Performance (RQ1)

To demonstrate the effectiveness of our model on reciprocal recommendation, we compare BAMBOO with all the baseline methods, and the results are shown in Table III. From the results, we can get the following observations:

- 1) The performance of our model surpasses all of the baseline methods on different evaluation metrics on two different sides. Furthermore, our model surpasses all the other models significantly with p-values smaller than 0.01 on evaluation metrics about matching prediction. This clearly proves the effectiveness of our BAMBOO model. Indeed, there are two primary reasons our model excels in comparison to other methods. First, we introduced two distinct modules for user modeling, respectively the Multi-behavior Sequence Encoder and the Heterogeneous Graph Encoder. With the aid of contrastive learning, we seamlessly merged the representations from both modules. This approach allows our model to effectively model the multi-behaviors of job seekers and recruiters in the recruitment market from both the expectation and competitiveness perspectives. Second, we proposed a training strategy based on multi-task learning, enabling our model to capture the bidirectional behaviors in the recruitment market, thereby achieving enhanced performance.
- 2) The DIEN model is significantly superior to the DIN model, which demonstrates the necessity of capturing the interest evolution of users in online recruitment platforms. Furthermore, the MBDIT and MBGRU models can achieve better performance than the DIEN model, which proves the effectiveness of modeling the multi-behaviors from the expectation perspective.
- 3) The HGT model can perform better than the DIN model, which proves the effectiveness of modeling the multi-behaviors from the competitiveness perspective. Furthermore, the HGT model trained on our bilateral heterogeneous interaction graph performs better than the DPGNN model. The rationality behind this is that the graph construction method in the DPGNN model converts the user interactions into a homogeneous graph, which limits the ability of models to capture the multi-typed user behaviors. This demonstrates the superiority of our method to convert user behaviors into a bilateral heterogeneous interaction graph.

F. Ablation Study (RQ1, RQ2)

To further validate the effectiveness of each component of our model, we also designed some simplified variants of our model as follows:

- *BAMBOO-Seq*: This method is a simplified variant of BAMBOO which only saves the sequence encoder part for behavior modeling.
- *BAMBOO-Seq-Apply*: This method is a simplified variant of BAMBOO-Seq which only train the model by the loss function on application prediction.

TABLE III
THE PERFORMANCE OF EACH MODEL

		Job Seeker as The Active Party				Recruiter as The Active Party			
		AUC-Match (↑)	LL-Match (↓)	AUC-Apply (↑)	LL-Apply (↓)	AUC-Match (↑)	LL-Match (↓)	AUC-Apply (↑)	LL-Apply (↓)
Arts	DIN	0.5995(.0034)	0.4614(.0020)	0.6496(.0035)	0.6326(.0027)	0.6353(.0024)	0.3980(.0018)	0.6276(.0077)	0.5842(.0033)
	DIEN	0.6156(.0033)	0.4566(.0012)	0.6896(.0061)	0.6116(.0029)	0.6463(.0019)	0.3942(.0012)	0.6055(.0066)	0.5892(.0020)
	HAN	0.5999(.0104)	0.4595(.0027)	0.6310(.0041)	0.6379(.0023)	0.6382(.0069)	0.3959(.0020)	0.5856(.0062)	0.5930(.0015)
	HGT	0.6556(.0030)	0.4476(.0025)	0.6902(.0018)	0.6107(.0024)	0.6633(.0021)	0.3900(.0019)	0.6896(.0037)	0.5568(.0040)
	MBGCN	0.6322(.0025)	0.4545(.0028)	0.6688(.0016)	0.6218(.0047)	0.6552(.0023)	0.3927(.0016)	0.6448(.0028)	0.5754(.0027)
	MBDIT	0.6502(.0011)	0.4478(.0011)	0.7510(.0011)	0.5673(.0025)	0.6849(.0010)	0.3836(.0010)	0.7563(.0009)	0.5119(.0015)
	MBGRU	0.6537(.0014)	0.4465(.0007)	0.7523(.0009)	0.5681(.0038)	0.6856(.0012)	0.3830(.0006)	0.7558(.0009)	0.5133(.0019)
	PJFNN	0.5917(.0034)	0.4623(.0012)	0.6260(.0030)	0.6412(.0032)	0.6347(.0024)	0.3968(.0007)	0.5984(.0034)	0.5917(.0009)
	DPGNN	0.6018(.0041)	0.4631(.0034)	0.6352(.0034)	0.6380(.0024)	0.6379(.0021)	0.3977(.0029)	0.6031(.0052)	0.5913(.0021)
	BOSS	0.6040(.0037)	0.4613(.0030)	0.6425(.0054)	0.6387(.0051)	0.6370(.0027)	0.3985(.0020)	0.6212(.0042)	0.5846(.0023)
	BAMBOO	0.6810(.0027)	0.4394(.0015)	0.7542(.0012)	0.5658(.0041)	0.6942(.0021)	0.3801(.0007)	0.7576(.0008)	0.5101(.0015)
Manu	DIN	0.7447(.0016)	0.4852(.0017)	0.6620(.0027)	0.6452(.0018)	0.6153(.0037)	0.3646(.0006)	0.6784(.0018)	0.4556(.0011)
	DIEN	0.7520(.0022)	0.4799(.0017)	0.6950(.0048)	0.6287(.0034)	0.6276(.0026)	0.3621(.0007)	0.6753(.0017)	0.4580(.0016)
	HAN	0.7369(.0017)	0.4896(.0019)	0.6507(.0036)	0.6498(.0030)	0.6132(.0031)	0.3644(.0010)	0.6761(.0020)	0.4563(.0007)
	HGT	0.7734(.0019)	0.4653(.0012)	0.6824(.0016)	0.6342(.0015)	0.6325(.0032)	0.3618(.0011)	0.7261(.0023)	0.4375(.0032)
	MBGCN	0.7639(.0020)	0.4723(.0016)	0.6746(.0025)	0.6386(.0022)	0.6302(.0026)	0.3626(.0017)	0.7049(.0026)	0.4450(.0013)
	MBDIT	0.7651(.0012)	0.4691(.0013)	0.7312(.0018)	0.6006(.0023)	0.6774(.0018)	0.3512(.0009)	0.7735(.0011)	0.4072(.0009)
	MBGRU	0.7657(.0011)	0.4688(.0013)	0.7324(.0012)	0.5986(.0017)	0.6792(.0011)	0.3505(.0013)	0.7763(.0009)	0.4055(.0006)
	PJFNN	0.7408(.0012)	0.4873(.0015)	0.6534(.0036)	0.6490(.0023)	0.6122(.0023)	0.3650(.0007)	0.6705(.0021)	0.4587(.0011)
	DPGNN	0.7496(.0017)	0.4820(.0042)	0.6545(.0013)	0.6482(.0029)	0.6154(.0018)	0.3653(.0014)	0.6741(.0035)	0.4579(.0021)
	BOSS	0.7472(.0023)	0.4844(.0027)	0.6592(.0021)	0.6516(.0042)	0.6142(.0022)	0.3663(.0021)	0.6778(.0038)	0.4577(.0025)
	BAMBOO	0.7893(.0012)	0.4523(.0017)	0.7335(.0012)	0.5975(.0011)	0.6840(.0010)	0.3491(.0013)	0.7774(.0007)	0.4051(.0014)
Serv	DIN	0.6796(.0022)	0.3958(.0016)	0.6137(.0015)	0.6530(.0018)	0.6264(.0024)	0.2155(.0009)	0.6912(.0023)	0.4900(.0016)
	DIEN	0.6955(.0052)	0.3900(.0018)	0.6621(.0086)	0.6328(.0035)	0.6366(.0027)	0.2144(.0004)	0.7109(.0045)	0.4806(.0032)
	HAN	0.6757(.0029)	0.3966(.0010)	0.6001(.0028)	0.6573(.0015)	0.6310(.0020)	0.2149(.0008)	0.6745(.0084)	0.4979(.0038)
	HGT	0.7194(.0015)	0.3819(.0009)	0.6462(.0019)	0.6402(.0010)	0.6614(.0016)	0.2115(.0005)	0.7274(.0013)	0.4727(.0013)
	MBGCN	0.7097(.0021)	0.3863(.0019)	0.6328(.0018)	0.6468(.0021)	0.6526(.0024)	0.2126(.0005)	0.7020(.0042)	0.4867(.0029)
	MBDIT	0.7262(.0018)	0.3788(.0011)	0.6976(.0008)	0.6116(.0008)	0.6969(.0013)	0.2066(.0006)	0.7685(.0008)	0.4474(.0012)
	MBGRU	0.7276(.0013)	0.3781(.0011)	0.6978(.0013)	0.6117(.0009)	0.6992(.0007)	0.2061(.0004)	0.7699(.0005)	0.4450(.0008)
	PJFNN	0.6763(.0028)	0.3970(.0017)	0.6008(.0026)	0.6588(.0025)	0.6241(.0026)	0.2154(.0003)	0.6693(.0025)	0.4978(.0013)
	DPGNN	0.6867(.0022)	0.3945(.0017)	0.6058(.0024)	0.6575(.0015)	0.6437(.0027)	0.2142(.0007)	0.6738(.0022)	0.4973(.0020)
	BOSS	0.6844(.0036)	0.3938(.0018)	0.6130(.0027)	0.6545(.0029)	0.6326(.0046)	0.2146(.0007)	0.6847(.0027)	0.4939(.0053)
	BAMBOO	0.7453(.0017)	0.3725(.0034)	0.7010(.0018)	0.6099(.0014)	0.7061(.0012)	0.2053(.0007)	0.7726(.0005)	0.4444(.0014)
Tech	DIN	0.6637(.0021)	0.3556(.0011)	0.6903(.0029)	0.5624(.0027)	0.6541(.0028)	0.4512(.0030)	0.6880(.0024)	0.4763(.0032)
	DIEN	0.6827(.0034)	0.3501(.0027)	0.7408(.0071)	0.5326(.0055)	0.6693(.0016)	0.4460(.0026)	0.7532(.0106)	0.4451(.0080)
	HAN	0.6773(.0066)	0.6666(.0010)	0.5688(.0030)	0.3546(.0015)	0.6657(.0046)	0.6526(.0031)	0.4824(.0015)	0.4518(.0027)
	HGT	0.6666(.0010)	0.3546(.0015)	0.6773(.0066)	0.5688(.0030)	0.6526(.0031)	0.4518(.0027)	0.6657(.0046)	0.4824(.0015)
	MBGCN	0.7029(.0022)	0.3448(.0017)	0.7086(.0022)	0.5542(.0035)	0.6757(.0035)	0.4468(.0052)	0.7244(.0022)	0.4591(.0030)
	MBDIT	0.6872(.0014)	0.3468(.0023)	0.7618(.0017)	0.5142(.0039)	0.6947(.0033)	0.4324(.0037)	0.7827(.0018)	0.4185(.0040)
	MBGRU	0.6885(.0017)	0.3453(.0018)	0.7633(.0015)	0.5109(.0036)	0.6964(.0027)	0.4319(.0029)	0.7857(.0024)	0.4146(.0031)
	PJFNN	0.6605(.0014)	0.3544(.0015)	0.6764(.0027)	0.5689(.0029)	0.6567(.0015)	0.4471(.0016)	0.6464(.0031)	0.4846(.0021)
	DPGNN	0.6929(.0028)	0.3459(.0019)	0.6880(.0021)	0.5611(.0043)	0.6715(.0015)	0.4448(.0016)	0.6547(.0027)	0.4842(.0036)
	BOSS	0.6745(.0020)	0.3519(.0026)	0.6916(.0025)	0.5599(.0025)	0.6639(.0023)	0.4446(.0018)	0.6836(.0022)	0.4718(.0033)
	BAMBOO	0.7396(.0023)	0.3283(.0013)	0.7730(.0012)	0.5040(.0026)	0.7164(.0022)	0.4238(.0025)	0.7919(.0020)	0.4085(.0019)

Bold face indicates the best result in terms of the corresponding metric.

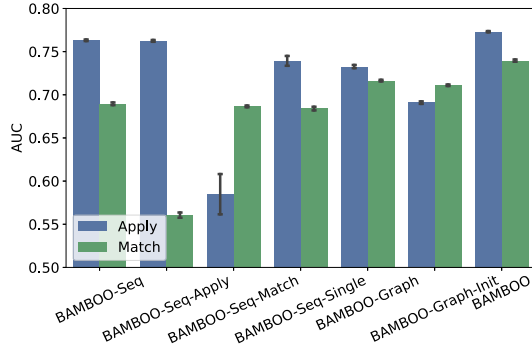
- *BAMBOO-Seq-Match*: This method is a simplified variant of BAMBOO-Seq which only train the model by the loss function on matching prediction.
- *BAMBOO-Seq-Single*: This method deletes the multi-behavior label embedding in the sequence encoder, and this method also deletes the sample in sequences with statu labels equal to 0.
- *BAMBOO-Graph*: This method is a simplified variant of BAMBOO which only saves the graph encoder part for behavior modeling.
- *BAMBOO-Graph-Init*: This method uses the initial interaction graph in Fig. 3 instead of the bilateral interaction graph in BAMBOO-Graph.

It is noteworthy that we designed BAMBOO-Seq, BAMBOO-Seq-Single, BAMBOO-Graph, and BAMBOO-Graph-Init to validate both the overall impact and individual contributions of our two primary modules, namely the Multi-behavior Sequence Encoder in Section IV-B and the Heterogeneous Graph Encoder in Section IV-C. Additionally, we introduced BAMBOO-Seq-Apply and BAMBOO-Seq-Match to verify the effects of the multiple loss functions presented in Section IV-E. The results of the ablation study are shown in Fig. 5, from which we can draw the following conclusions:

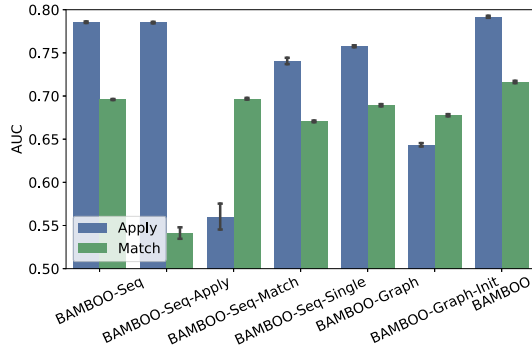
- 1) Our BAMBOO model performs significantly better than BAMBOO-Seq and BAMBOO-Graph, which demonstrates the necessity of modeling the bilateral multi-behaviors from two different perspectives.
- 2) The BAMBOO-Seq model performs better than BAMBOO-Seq-Apply and BAMBOO-Seq-Match, which proves the effectiveness of our multi-task learning strategy.
- 3) The BAMBOO-Seq model outperforms BAMBOO-Seq-Single by a large margin, which verifies the effectiveness of our multi-behavior sequential modeling method.
- 4) The BAMBOO-Graph model performs better than the BAMBOO-Graph-Init, which demonstrates the effectiveness of our bilateral heterogeneous interaction graph constructing method.

G. Parameter Sensitivity (RQ2)

We evaluated how hyper-parameter α affected the performance of our model in this section, and the results are shown in Fig. 6. We can find that BAMBOO performs best when we set $\alpha = 0.98$, which verifies the effectiveness of the contrastive learning loss. Furthermore, we can observe that the performance

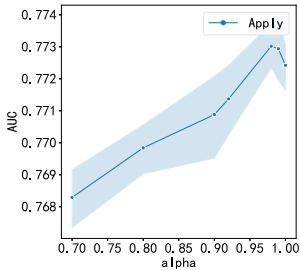


(a) The performance of different models on the job seeker side.

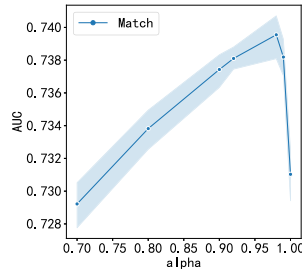


(b) The performance of different models on the recruiter side.

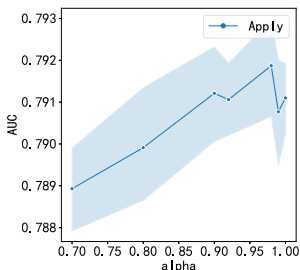
Fig. 5. The results of ablation study.



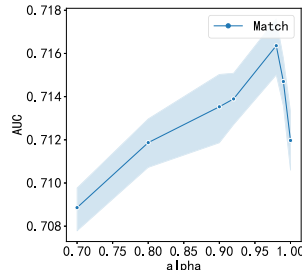
(a) The application prediction performance of BAMBOO on the job seeker side.



(b) The matching prediction performance of BAMBOO on the job seeker side.



(c) The application prediction performance of BAMBOO on the recruiter side.



(d) The matching prediction performance of BAMBOO on the recruiter side.

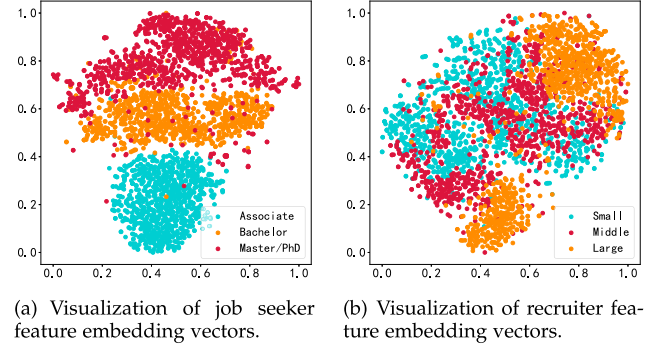
 Fig. 6. The performance of BAMBOO with different α .


Fig. 7. Visualization of feature embedding vectors.

of BAMBOO is good enough when α ranges from 0.8 to 0.99, which proves the robustness of the contrastive learning method.

H. Case Study (RQ3)

In this part, we present some cases to illustrate the effectiveness of our model and reveal some interesting rules based on the derived insights on online recruitment.

1) *Case Study on Feature Embedding*: As we mentioned before, different job seekers and recruiters have different features, and we converted these features to embedding vectors for further calculation. To analyze the influence of different features for the embedding vectors, we randomly selected 3,000 job seekers and 3,000 recruiters, and projected their corresponding feature embedding vectors into two-dimensional space with t-SNE [46]. The results are shown in Fig. 7. Specifically, Fig. 7(a) shows the visualization results of job seekers with different education degrees, respectively "Associate Degree", "Bachelor Degree", and "Master/PhD Degree". We can find that the embedding vectors of these job seekers are grouped into distinct clusters, which indicates that we can clearly distinguish the target recruiters of different job seekers based on their educational degrees. Furthermore, Fig. 7(b) shows the visualization results of recruiters with different company scales, respectively "Small", "Middle", and "Large". We can observe that the embedding vectors of these recruiters do not exhibit significant distinctions, which indicates that the scale of a company does not have a decisive impact on the decisions of job seekers.

2) *Case Study on Model Output and Explainability*: Next, We investigated how the profile features of job seekers and recruiters influence the model output, thereby delving deeper into the explainability of our model. First, we selected three job seeker-recruiter pairs from the technology dataset. The recruiters three recruiters are all large companies in big cities, and posted job positions related to front-end engineers, while the main difference between the three job seekers is their educational backgrounds. The results are shown in Table IV. We can clearly see that the higher the degree of job seekers, the greater the probability of being accepted by recruiters. At the same time, recruiters tend to invite candidates with high educational degrees actively, but the conversion rate of recruiting high-educated

TABLE IV
THE OUTPUT RESULTS FROM THE MULTI-TASK OUTPUT LAYER OF BAMBOO

Job Seeker Features	Recruiter Features	$p_{apply}^{s \rightarrow r}$	$p_{acc}^{s \rightarrow r}$	$p_{apply}^{r \rightarrow s}$	$p_{acc}^{r \rightarrow s}$
Degree: Master, Major: Computer Science Expect Salary: 15-30K	Scale: Large, City: Shenzhen, Salary: 15-17K Position: Front-end Engineer	0.53	0.74	0.97	0.07
Degree: Bachelor, Major: Computer Science Expect Salary: 11-16K	Scale: Large, City: Beijing, Salary: 13-22K Position: Front-end Engineer	0.82	0.35	0.60	0.57
Degree: Associate, Major: Software Engineering Expect Salary: 11-16K	Scale: Large, City: Shenzhen, Salary: 10-12K Position: Front-end Engineer	0.70	0.19	0.40	0.62

TABLE V
THE APPLICATION PROBABILITY COMPARISON RESULTS

Job Seeker Features	Recruiter Features	$p_{apply}^{s \rightarrow r}$	$\hat{p}_{apply}^{s \rightarrow r}$
Degree: Bachelor Major: Mining Engineering Expect Salary: 16-17K	Scale: Large Position: Architect Salary: 30-40K	0.55	0.66

TABLE VI
THE MATCHING PROBABILITY COMPARISON RESULTS

Job Seeker Features	Recruiter Features	$p_{apply}^{s \rightarrow r}$	$\hat{p}_{apply}^{s \rightarrow r}$
Degree: Master Major: Computer Science Expect Salary: 15-23K	Scale: Large Position: System Engineer Salary: 12-24K	0.87	0.74

talents is also lower. This phenomenon is consistent with our common sense. Additionally, we observed that the model only predicts higher application and matching probabilities when the features of both job seekers and recruiters align well. This indicates that our model has accurately learned the interaction between features, thereby providing support for the explainability. To further validate the impact of different features on the model output, thereby providing a basis for the model explainability, we conducted a statistical analysis on our dataset. Specifically, we calculated the mean value of $p_{acc}^{s \rightarrow r}$ corresponding to job seekers with bachelor degree and master degree. The results are 27.59% and 30.64%, which means job seekers with a master degree have a relative increase of 11% in job search success rate compared to job seekers with a bachelor degree.

3) *Case Study on Multi-Behavior Label*: In this part, we studied the influence of the multi-behavior labels via simulation experiments on two cases. We denote the job seekers and recruiters in two cases as s_1 , s_2 , r_1 , and r_2 , respectively.

Case 1: As shown in Table V, the original estimated probability $p_{apply}^{s \rightarrow r}$ that the job seeker s_1 will apply for the job posted by r_1 is relatively lower since the salary of this job is much higher than his expected salary. However, if we converted all the status labels equal to 1 to 2, i.e., assuming that all the applications in his historical records were successful, the re-calculated probability would be higher than the original probability. This may suggest that if the job seeker has a smooth job hunting experience with many successful applications, it will greatly encourage the job seeker to apply for this job.

Case 2: As shown in Table VI, the conditions of the job posted by r_2 are appealing to s_2 and the model predicts that he has a large probability to apply for it. However, if we converted all the status labels equal to 2 in the historical record of s_1 to 1, i.e., assuming that all the historical applications were failed, the estimated probability will drop to 0.74 from 0.87. This may indicate that if the candidate has experienced too many

failures before, it will affect his/her confidence and reduce the probability of application.

The above simulation experiments demonstrate the effectiveness of our model to capture the influence of the multi-behavior interactions between different job seekers and recruiters. Additionally, the aforementioned experiments provide a foundation for the explainability of our model in practical applications, such as explaining why the recommendation list changes after users exhibit different behaviors within the system, and further analysis like the prediction of skill popularity and market activity [47], [48].

I. Online A/B Test

To further validate the effectiveness of our proposed BAMBOO model, we conducted a series of online A/B tests in our online recruitment platform on both the job seeker side and the recruiter side. On the job seeker side, for the control group, 3% of job seekers are randomly selected and presented with the recommendation probability $p_{match}^{s \rightarrow r}$ generated by the current online Base model. For the experimental group, 3% of job seekers are presented with recommendation probability generated by our BAMBOO model. Note that Within this platform, the maximum traffic allowed for A/B testing corresponding to a single model is 3%, and we chose this maximum traffic to test the models. To ensure the fairness, reliability, and accuracy of the online A/B test results, during the testing, the job seekers and recruiters for both the control group and the experimental group were randomly selected by the platform, with no statistically significant differences in their profile features. The above models are deployed online simultaneously for four days. On the recruiter side, the settings are almost the same and the recruiters are presented with the recommendation probability $p_{match}^{r \rightarrow s}$. The above models are deployed online simultaneously for three days. Moreover, we adopt two metrics, respectively Realistic Apply Rate (RAR) and Realistic Match Rate (RMR), to compare the online performance of the models as follows:

$$RAR = \frac{\#applies}{\#impressions}, \quad RMR = \frac{\#matches}{\#impressions}. \quad (31)$$

The results of online tests are shown in Fig. 8. From the results we can find that the BAMBOO model performs significantly better than the Base model. The results clearly demonstrate the effectiveness of our multi-type behavior modeling and multi-task learning methods.

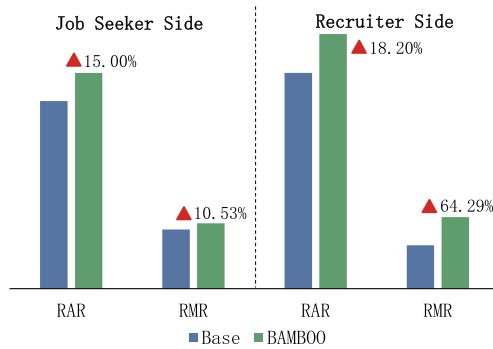


Fig. 8. The results of online A/B test. Due to the commercial privacy, we do not report precise metric numbers here, but only relative improvement values.

VI. CONCLUSION

In this paper, we studied the reciprocal recommendation problem in online recruitment. We proposed a reciprocal recommendation model named BAMBOO which can model the bilateral multi-behaviors between job seekers and recruiters from two different perspectives, respectively expectation perspective and competitiveness perspective. Specifically, for the expectation perspective, we proposed to build bilateral multi-behavior sequences, and we extracted the sequence embedding by a transformer-based model. For the competitiveness perspective, we proposed to construct a bilateral interaction heterogeneous graph between the job seekers and recruiters, and we leveraged a heterogeneous graph transformer-based model for graph encoding. We further proposed to decompose the matching probability into the product of two different parts, and train our model based on a multi-task learning loss and a contrastive learning loss. Finally, we evaluated the performance of our model through both offline and online testing, and the results clearly validated the effectiveness of the BAMBOO model.

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